ECE 102 Homework 1

Edi Zhang

April 12, 2025

1. Sketch and write the even and odd components of the following signal:

$$x(t) = \text{relu}(t) = \begin{cases} t & t \geq 0 \\ 0 & t < 0 \end{cases} \tag{1}$$

Solution:

Recall that

$$x_e(t) \triangleq \frac{1}{2}(x(t) + x(-t)) \quad \text{and} \quad x_o(t) \triangleq \frac{1}{2}(x(t) - x(-t)) \tag{2}$$

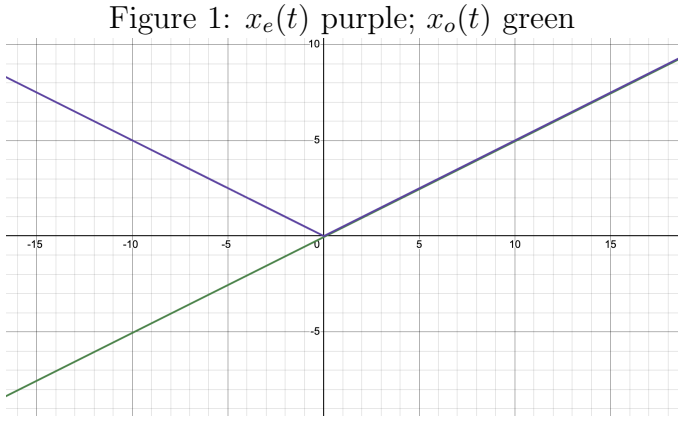
Thus

$$x_e(t) = \begin{cases} \frac{1}{2}t & t \geq 0 \\ -\frac{1}{2}t & t < 0 \end{cases} \tag{3}$$

and

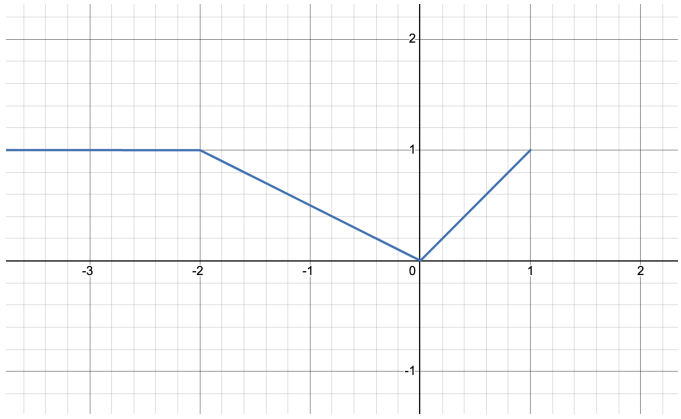
$$x_o(t) = \frac{1}{2}t \tag{4}$$

Figure 1: $x_e(t)$ purple; $x_o(t)$ green



2. **Time scaling and shifting**

   (a) Consider the following signal Sketch the following:



i. $x(\frac{1}{2}t - 1) = x(\frac{1}{2}(t - 2))$

ii. $x(-3t - 2) = x(-3(t + \frac{2}{3}))$

Solution:

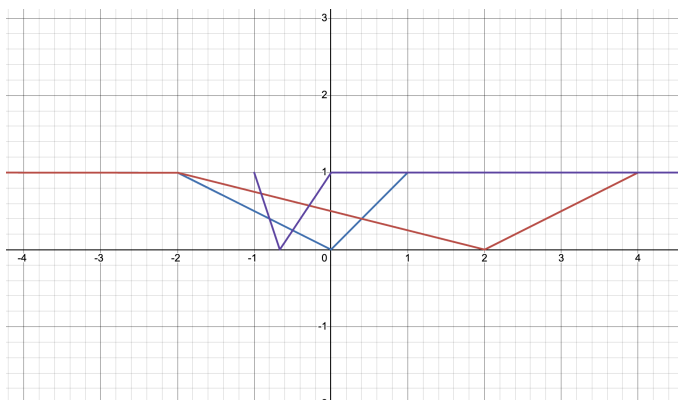


Figure 2: Red: (i), Purple: (ii)

(b) The figure below shows two signals $z(t)$ and $y(t)$. Can you express $y(t)$ in terms of $z(t)$

Solution:

$$y(t) = z\left(-2\left(t + 1\right)\right) \tag{5}$$

3. **Periodic signals**

(a) Determine periodic or not. If periodic, determine fundamental period and frequency

i. $x_1(t) = \sin(2t + \pi/3)$

Solution:

$$\begin{aligned} x_1(t) &= x_1(t + kT_0) && (6)\\ &= \sin(2(t + kT_0) + \pi/3) && (7)\\ &= \sin(2t + 2kT_0 + \pi/3) && (8) \end{aligned}$$

Solve

$$2\pi k = 2kT_0 \implies T_0 = \pi \quad f = \frac{1}{T_0} = \frac{1}{\pi} \tag{9}$$

ii. $x_2(t) = \cos(\sqrt{2}\pi t)$

Solution:

$$\begin{aligned} x_2(t) &= x_2(t + kT_0) && (10)\\ &= \cos(\sqrt{2}\pi(t + kT_0)) && (11)\\ &= \cos(\sqrt{2}\pi t + \sqrt{2}\pi kT_0)) && (12) \end{aligned}$$

Solve

$$2\pi k = \sqrt{2}\pi kT_0 \implies T_0 = \sqrt{2} \quad f = \frac{1}{\sqrt{2}} \tag{13}$$

iii. $x_3(t) = \sin^2(3\pi t + 3)$

Solution:

$$\begin{aligned} x_3(t) &= x_3(t + kT_0) && (14)\\ &= \sin^2(3\pi(t + kT_0) + 3) && (15)\\ &= \sin^2(3\pi t + 3\pi kT_0 + 3) && (16) \end{aligned}$$

Solve

$$\pi k = 3\pi kT_0 \implies T_0 = \frac{1}{3} \quad f = 3 \tag{17}$$

(Fundamental period of $\sin^2(x)$ is π)

iv. $x_4(t) = x_1(t) + x_2(t)$

Solution:

$$\begin{aligned} x_4(t) &= x_4(t + kT_0) && (18)\\ &= x_1(t + kT_0) + x_2(t + kT_0) && (19)\\ &= \sin(2(t + kT_0) + \pi/3) + \cos(\sqrt{2}\pi(t + kT_0)) && (20)\\ &= \sin(2t + \pi/3 + 2kT_0) + \cos(\sqrt{2}\pi t + \sqrt{2}\pi kT_0) && (21) \end{aligned}$$

$T_0^{(1)} = \pi, \quad T_0^{(2)} = \sqrt{2}.$

The fundamental period is $\text{lcm}(\pi, \sqrt{2})$. Choose $m, n \in \mathbb{Z}$ such that $m\pi = n\sqrt{2}$

$$\frac{m}{n} = \frac{\sqrt{2}}{\pi} \tag{22}$$

However, $\frac{\sqrt{2}}{\pi}$ is irrational, and therefore there does not exist m, n that satisfy this expression and thus there is not fundamental period.

v. $x_5(t) = x_1(\pi t) + x_3(t)$

Solution:

$$\begin{aligned} x_5(t) &= x_5(t + kT_0) && (23)\\ &= x_1(\pi(t + kT_0)) + x_3(t + kT_0) && (24)\\ &= \sin(2(\pi(t + kT_0)) + \pi/3) + \sin^2(3\pi(t + kT_0) + 3) && (25)\\ &= \sin(2\pi t + 2\pi kT_0 + \pi/3) + \sin^2(3\pi t + 3\pi kT_0 + 3) && (26) \end{aligned}$$

Solving,

$$2\pi k = 2\pi kT_0^{(1)} \implies T_0^{(1)} = 1 \tag{27}$$

and

$$\pi k = 3\pi kT_0^{(3)} \implies T_0^{(3)} = \frac{1}{3} \tag{28}$$

$\text{lcm}(1, \frac{2}{3}) = 2$. Thus $T_0 = 1$, and $f = 1$

vi. $x_6(t) = e^{-t}x_1(t)$

Solution:

$$\begin{aligned} x_6(t) &= x_6(t + kT_0) && (29)\\ &= e^{-(t+kT_0)}x_1(t + kT_0) && (30)\\ &= e^{-(t+kT_0)}\sin(2(t + kT_0) + \pi/3) && (31)\\ &= e^{-(t+kT_0)}\sin(2t + 2kT_0 + \pi/3) && (32)\\ &= e^{-kT_0}e^{-t}\sin(2t + 2kT_0 + \pi/3) && (33) \end{aligned}$$

Therefore we have

$$e^{-kT_0} = 1 \quad \text{and} \quad 2kT_0 = 2\pi k \tag{34}$$

which has no solution. Thus the signal is not periodic

vii. $x_7(t) = e^{j(\pi t+1)}x_2(t)$

Solution:

$$\begin{aligned} x_7(t) &= x_7(t + kT_0) && (35)\\ &= e^{j(\pi(t+kT_0)+1)}\cos(\sqrt{2}\pi(t + kT_0)) && (36)\\ &= e^{j(\pi t+\pi kT_0+1)}\cos(\sqrt{2}\pi t + \sqrt{2}\pi kT_0)) && (37)\\ &= e^{j\pi kT_0}e^{j(\pi t+1)}\cos(\sqrt{2}\pi t + \sqrt{2}\pi kT_0)) && (38) \end{aligned}$$

Therefore we have

$$e^{j\pi kT_0} = 1 \quad \text{and} \quad \sqrt{2}\pi kT_0 = 2\pi k \tag{39}$$

When $j = 0$, we have $T_0 = \sqrt{2}$ and $f = \frac{1}{\sqrt{2}}$. When $j \neq 0$, the system has no solution and the signal is not periodic.

(b) Assume that the signal $x(t)$ is periodic with period T_0, and that $x(t)$ is odd. What is the value of $x(T_0)$?

Solution: Given $x(t) = x(t + T_0)$ and $x(t) = -x(-t)$,

$$\begin{aligned} x(T_0) &= -x(-T_0) && (40)\\ &= -x(-T_0 + T_0) && (41)\\ &= -x(0) && (42) \end{aligned}$$

Since $x(t)$ is an odd function

$$x(0) = -x(-0) = -x(0) \tag{43}$$

This is only possible if $x(0) = 0$. Therefore $x(T_0) = -x(0) = 0$

(c) If $x(t)$ is periodic, are the even and odd components of $x(t)$ also periodic?

Solution:

Let $x(t)$ be periodic with some fundamental period T_0, then

$$x(t) = x(t + T_0) \tag{44}$$

Recall that

$$x_e(t) \triangleq \frac{1}{2}(x(t) + x(-t)) \quad \text{and} \quad x_o(t) \triangleq \frac{1}{2}(x(t) - x(-t)) \tag{45}$$

Thus

$$\begin{aligned} x_e(t + T_0) &= \frac{1}{2}(x(t + T_0) + x(T_0 - t)) && (46)\\ &= \frac{1}{2}(x(t) + x(-t)) && (47)\\ &= x_e(t) && (48) \end{aligned}$$

and

$$\begin{aligned} x_o(t + T_0) &= \frac{1}{2}(x(t + T_0) - x(T_0 - t)) && (49)\\ &= \frac{1}{2}(x(t) - x(-t)) && (50)\\ &= x_o(t) && (51) \end{aligned}$$

The even and odd components of $x(t)$ are both periodic.

4. **Energy and power signals**

(a) Determine whether the following signals are energy or power signals. If the signal is an energy signal, determine its energy. If the signal is a power signal, determine its power.

i. $x(t) = e^{-|t|}$

Solution:

Since the area under the curve is bounded, this is an energy signal.

$$E_x = \lim_{T\to\infty} \int_{-T}^{T} |x(t)|^2 dt \tag{52}$$

$$= \lim_{T\to\infty} \int_{-T}^{T} |e^{-|t|}|^2 dt \tag{53}$$

$$= \lim_{T\to\infty} \int_{-T}^{T} e^{-2|t|} dt \tag{54}$$

$$= \lim_{T\to\infty} \int_{-T}^{0} e^{-2|t|} dt + \lim_{T\to\infty} \int_{0}^{T} e^{-2|t|} dt \tag{55}$$

Consider: $t \leq 0$, then $|t| = -t$

$$E_- = \lim_{T\to\infty} \int_{-T}^{0} e^{-2|t|} dt \tag{56}$$

$$= \lim_{T\to\infty} \left[\frac{1}{2}e^{2t}\right]_{-T}^{0} \tag{57}$$

$$= \lim_{T\to\infty} \left[\frac{1}{2}e^{0} - \frac{1}{2}e^{-2T}\right] \tag{58}$$

$$= \frac{1}{2} \tag{59}$$

Since the function is symmetrical on both sides, $x(t)$ is an energy signal with energy $E_x = 2 \cdot \frac{1}{2} = 1$

ii. $x(t) = \begin{cases} \frac{1}{\sqrt{t}}, & \text{if } t \geq 1 \\ 0, & \text{otherwise} \end{cases}$

Solution:

Since the function squared is unbounded (harmonic series), it has infinite energy.

Consider the power

$$
\begin{aligned}
P_x &= \lim_{T\to\infty} \frac{1}{2T}\int_1^T \frac{1}{t}dt && (60)\\
&= \lim_{T\to\infty} \frac{1}{2T}[\ln t]_1^T && (61)\\
&= \lim_{T\to\infty} \frac{\ln T}{2T} && (62)\\
&= \lim_{T\to\infty} \frac{\frac{1}{T}}{2} && (63)\\
&= 0 && (64)
\end{aligned}
$$

Thus the signal is neither an energy nor a power signal

iii. $x(t) = \begin{cases} 1+e^{-t}, & \text{if } t \geq 0 \\ 0, & \text{otherwise} \end{cases}$

Solution:

Since $\lim x(t) = 1$, it must have infinite energy and thus is not an energy signal. Consider the power

$$
\begin{aligned}
P_x &= \lim_{T\to\infty} \frac{1}{2T}\int_1^T (1+e^{-t})dt && (65)\\
&= \lim_{T\to\infty} \frac{1}{2T}[t-e^{-t}]_1^T && (66)\\
&= \lim_{T\to\infty} \frac{1}{2T}[T-e^{-T}-1+e^{-1}] && (67)\\
&= \lim_{T\to\infty} \frac{T-1+e^{-1}}{2T} && (68)\\
&= \frac{1}{2} && (69)
\end{aligned}
$$

$x(t)$ is a power signal with power $\frac{1}{2}$

(b) Show the following two properties

- If $x(t)$ is an even signal and $y(t)$ is an odd signal, then $x(t)y(t)$ is an odd signal;

  Solution:

  Given $x(t) = x(-t)$ and $y(t) = -y(-t)$, then $x(t)y(t) = -x(-t)y(-t)$, which means $x(t)y(t)$ is odd

- If $z(t)$ is an odd signal, then for any $\tau > 0$ we have:

$$\int_{-\tau}^{\tau} z(t)dt = 0 \qquad (70)$$

Solution:

Given $z(t) = -z(-t)$

$$\int_{-\tau}^{\tau} z(t)dt = \int_{-\tau}^{0} z(t)dt + \int_{0}^{\tau} z(t)dt \tag{71}$$
$$= \int_{0}^{\tau} -z(-t)dt + \int_{0}^{\tau} z(t)dt \tag{72}$$
$$= 0 \tag{73}$$

Use these two properties to show that the energy of $x(t)$ is the sum of the energy of its even component $x_e(t)$ and the energy of its odd component $x_0(t)$, i.e.,

$$E_x = E_{x_e} + E_{x_o} \tag{74}$$

Assume $x(t)$ is a real signal

Solution:

Recall that

$$x_e(t) \triangleq \frac{1}{2}(x(t) + x(-t)) \quad \text{and} \quad x_o(t) \triangleq \frac{1}{2}(x(t) - x(-t)) \tag{75}$$

Then

$$E_{x_e} + E_{x_0} = \lim_{T\to\infty} \int_{-T}^{T} |x_e(t)|^2 dt + \lim_{T\to\infty} \int_{-T}^{T} |x_o(t)|^2 dt \tag{76}$$
$$= \lim_{T\to\infty} \int_{-T}^{T} |x_e(t)|^2 + |x_o(t)|^2 dt \tag{77}$$
$$= \lim_{T\to\infty} \int_{-T}^{T} |x_e(t) + x_o(t)|^2 - x_e(t)x_o(t)dt \tag{78}$$
$$= \lim_{T\to\infty} \int_{-T}^{T} |x_e(t) + x_o(t)|^2 dt \tag{79}$$
$$= \lim_{T\to\infty} \int_{-T}^{T} |x(t)|^2 dt \tag{80}$$
$$= E_x \tag{81}$$

5. **Euler's identity and complex numbers**

   (a) Use Euler's formula to prove the following identities:

      i. $\cos^2(\theta) + \sin^2(\theta) = 1$

Solution:

$$\begin{aligned}
\cos^2(\theta) + \sin^2(\theta) &= (\cos(\theta) + i\sin(\theta))(\cos(\theta) - i\sin(\theta)) && (82)\\
&= e^{i\theta}e^{-i\theta} && (83)\\
&= e^0 && (84)\\
&= 1 && (85)
\end{aligned}$$

ii. $\cos(\theta + \psi) = \cos(\theta)\cos(\psi) - \sin(\theta)\sin(\psi)$

Solution: Note that

$$\cos\theta = \frac{1}{2}\left[e^{j\theta} + e^{-j\theta}\right] \quad \text{and} \quad \sin\theta = \frac{1}{2}\left[e^{j\theta} - e^{-j\theta}\right] \tag{86}$$

(RHS)

$$\begin{aligned}
&\cos(\theta)\cos(\psi) - \sin(\theta)\sin(\psi) && (87)\\
&= \frac{1}{4}\left(e^{j\theta} + e^{-j\theta}\right)\left(e^{j\psi} + e^{-j\psi}\right) + \frac{1}{4}\left(e^{j\theta} - e^{-j\theta}\right)\left(e^{j\psi} - e^{-j\psi}\right) && (88)\\
&= \frac{1}{4}\left[\left(e^{j\theta} + e^{-j\theta}\right)\left(e^{j\psi} + e^{-j\psi}\right) + \left(e^{j\theta} - e^{-j\theta}\right)\left(e^{j\psi} - e^{-j\psi}\right)\right] && (89)\\
&= \frac{1}{4}\left[2\left(e^{j(\theta+\psi)} + e^{-j(\theta+\psi)}\right)\right] && (90)\\
&= \frac{1}{2}\left[e^{j(\theta+\psi)} + e^{-j(\theta+\psi)}\right] && (91)\\
&= \cos(\theta + \psi) && (92)
\end{aligned}$$

(b) Show that $e^{j\theta} = 2\sin(\frac{\theta}{2})e^{j[(\theta+\pi)/2]} + 1$

Solution:

$$\begin{aligned}
&2\sin\left(\frac{\theta}{2}\right)e^{j[(\theta+\pi)/2]} + 1 && (93)\\
&\qquad = 2\sin\left(\frac{\theta}{2}\right)\left(\cos\left(\frac{\pi+\theta}{2}\right) + j\sin\left(\frac{\pi+\theta}{2}\right)\right) + 1 && (94)
\end{aligned}$$

Using $\cos(\theta+\psi) = \cos(\theta)\cos(\psi) - \sin(\theta)\sin(\psi)$, we get

$$= 2\sin\left(\frac{\theta}{2}\right)\left(-\sin\left(\frac{\theta}{2}\right) + j\cos\left(\frac{\theta}{2}\right)\right) + 1 \tag{95}$$

$$= 1 - 2\sin^2\left(\frac{\theta}{2}\right) + j2\sin\left(\frac{\theta}{2}\right)\cos\left(\frac{\theta}{2}\right) \tag{96}$$

$$= \cos(\theta) + j\sin(\theta) \tag{97}$$

$$= e^{j\theta} \tag{98}$$

(c) $x(t) = (5+\sqrt{2}j)e^{j(t+2)}$ and $y(t) = 1/(2-j)$

i. Compute the real and imaginary parts of $x(t)$ and $y(t)$.

Solution:

$$x(t) = (5+\sqrt{2}j)e^{j(t+2)} \tag{99}$$

$$= (5+\sqrt{2}j)(\cos(t+2) + j\sin(t+2)) \tag{100}$$

$$= 5\cos(t+2) + 5j\sin(t+2) + \sqrt{2}j\cos(t+2) - \sqrt{2}\sin(t+2) \tag{101}$$

$$= (5\cos(t+2) - \sqrt{2}\sin(t+2)) + (5\sin(t+2) + \sqrt{2}\cos(t+2))j \tag{102}$$

$\mathrm{Re}(x(t)) = 5\cos(t+2) - \sqrt{2}\sin(t+2), \quad \mathrm{Im}(x(t)) = 5\sin(t+2) + \sqrt{2}\cos(t+2)$

$$y(t) = 1/(2-j) \tag{103}$$

$$= \frac{1}{2-j}\cdot\frac{2+j}{2+j} \tag{104}$$

$$= \frac{2+j}{4-j^2} \tag{105}$$

$$= \frac{2+j}{5} \tag{106}$$

$$= \frac{2}{5} + \frac{1}{5}j \tag{107}$$

$\mathrm{Re}(y(t)) = \frac{2}{5}, \quad \mathrm{Im}(y(t)) = \frac{1}{5}$

ii. Compute the magnitude and phase of $x(t)$ and $y(t)$.

Solution: Magnitude:

$$|x(t)| = |(5+\sqrt{2}j)| \cdot |e^{j(t+2)}| \tag{108}$$

$$= \sqrt{5^2 + \sqrt{2}^2} \tag{109}$$

$$= \sqrt{27} \tag{110}$$

$$= 3\sqrt{3} \tag{111}$$

$$|y(t)| = \left|\frac{1}{2-j}\right| \tag{112}$$
$$= \frac{1}{|2-j|} \tag{113}$$
$$= \frac{1}{\sqrt{2^2+(-1)^2}} \tag{114}$$
$$= \frac{1}{\sqrt{5}} \tag{115}$$

Phase:

$$\arg x(t) = \arctan\left(\frac{5\sin(t+2)+\sqrt{2}\cos(t+2)}{5\cos(t+2)-\sqrt{2}\sin(t+2)}\right) \tag{116}$$
$$= \arg(5+\sqrt{2}j) + (t+2) \tag{117}$$
$$= \arctan\left(\frac{\sqrt{2}}{5}\right) + t + 2 \tag{118}$$

$$\arg\, y(t) = \arctan\frac{1/5}{2/5} = \arctan\frac{1}{2} \tag{119}$$